

GS 21/L Physical and Structural Geology

Module 7 **Volcanism**

COURSE OUTCOMES:

- At the end of this module, students will be able to:
- ✚ Analyze different surface geological features using basic physical geology principles.
- ✚ Classify different rock forming minerals and rock types typically found above and below the Earth's surface.

INTENDED LEARNING OUTCOMES:

- At the end of this module, the students will be able to:
- ✚ Discuss the different volcanic landforms, hazards and eruption styles.

PROGRAM OUTCOMES:

- ✚ Apply knowledge of mathematics and science to solve complex petroleum engineering problems;
- ✚ design and conduct experiments, as well as to analyze and interpret data;
- ✚ identify, formulate, and solve complex petroleum engineering problems;

TIME FRAME:

- ✚ This module can be covered in a week.

Volcanism

any of various processes and phenomena associated with the surficial discharge of molten rock, pyroclastic fragments, or hot water and steam, including volcanoes, geysers, and fumaroles. (<https://www.britannica.com/science/volcanism>)

VOLCANISM

THERE ARE THREE ERUPTION MECHANISMS:

1. **Magmatic eruptions** – a type of eruption wherein gas is released due to decompression. It involves the decompression of gas within magma that propels it forward
2. **Phreatomagmatic eruptions**- Thermal contraction from chilling on contact with water. It is driven by the compression of gas within magma, the direct opposite of the process powering magmatic activity.
3. **Phreatic eruptions**- Ejection of entrained particles during steam eruptions. It is driven by the [superheating](#) of [steam](#) via contact with [magma](#); these eruptive types often exhibit no magmatic release, instead causing the granulation of existing rock.



1. Magmatic Eruption

SUBTYPES OF MAGMATIC ERUPTION

A. Hawaiian eruption - is a type of volcanic eruption, named after the Hawaiian volcanoes with which this eruptive type is hallmark. Hawaiian eruptions are the calmest types of volcanic events, characterized by the effusive eruption of very fluid basalt-type lavas with low gaseous content. The volume of ejected material from Hawaiian eruptions is less than half of that found in other eruptive type.



1. Magmatic eruption

SUBTYPES OF MAGMATIC ERUPTION

B. STROMBOLIAN - a type of volcanic eruption, named after the volcano Stromboli, which has been erupting continuously for centuries. Strombolian eruptions are driven by the bursting of gas bubbles within the magma. These gas bubbles within the magma accumulate and coalesce into large bubbles, called gas slugs. These grow large enough to rise through the lava column. Upon reaching the surface, the difference in air pressure causes the bubble to burst with a loud pop, throwing magma in the air in a way similar to a soap bubble. Because of the high gas pressures associated with the lavas, continued activity is generally in the form of episodic explosive eruptions accompanied by the distinctive loud blasts. During eruptions, these blasts occur as often as every few minutes.

1. Magmatic eruption

SUBTYPES OF MAGMATIC ERUPTION

C. VULCANIAN - type of volcanic eruption, named after the volcano Vulcano. It was named so following Giuseppe Mercalli's observations of its 1888–1890 eruptions. In Vulcanian eruptions, intermediate viscous magma within the volcano make it difficult for vesiculate gases to escape. Similar to Strombolian eruptions, this leads to the buildup of high gas pressure, eventually popping the cap holding the magma down and resulting in an explosive eruption. However, unlike Strombolian eruptions, ejected lava fragments are not aerodynamic; this is due to the higher viscosity of Vulcanian magma and the greater incorporation of crystalline material broken off from the former cap. They are also more explosive than their Strombolian counterparts, with eruptive columns often reaching between 5 and 10 km (3 and 6 mi) high. Lastly, Vulcanian deposits are andesitic to dacitic rather than basaltic.



1. Magmatic eruption

SUBTYPES OF MAGMATIC ERUPTION

d. PELEAN- are a type of volcanic eruption, named after the volcano Mount Pelée in Martinique, the site of a massive Peléan eruption in 1902 that is one of the worst natural disasters in history. In Peléan eruptions, a large amount of gas, dust, ash, and lava fragments are blown out the volcano's central crater, driven by the collapse of rhyolite, dacite, and andesite lava dome collapses that often create large eruptive columns. An early sign of a coming eruption is the growth of a so-called Peléan or lava spine, a bulge in the volcano's summit preempting its total collapse. The material collapses upon itself, forming a fast-moving pyroclastic flow (known as a block-and-ash flow) that moves down the side of the mountain at tremendous speeds, often over 150 km (93 mi) per hour. These massive landslides make Peléan eruptions one of the most dangerous in the world, capable of tearing through populated areas and causing massive loss of life. The 1902 eruption of Mount Pelée caused tremendous destruction, killing more than 30,000 people and completely destroying the town of St. Pierre, the worst volcanic event in the 20th century.



PELEAN ERUPTION

Volcanoes known to have Peléan activity include:

- ❑ **Mount Pelée, Martinique.** The 1902 eruption of Mount Pelée completely devastated the island, destroying the town of St. Pierre and leaving only 3 survivors. The eruption was directly preceded by lava dome growth.
- ❑ **Mayon Volcano**, the Philippines most active volcano. It has been the site of many different types of eruptions, Peléan included. Approximately 40 ravines radiate from the summit and provide pathways for frequent pyroclastic flows and mudslides to the lowlands below. Mayon's most violent eruption occurred in 1814 and was responsible for over 1200 deaths.
- ❑ **The 1951 Peléan eruption of Mount Lamington.** Prior to this eruption the peak had not even been recognized as a volcano. Over 3,000 people were killed, and it has become a benchmark for studying large Peléan eruptions



1. Magmatic eruption

SUBTYPES OF MAGMATIC ERUPTION

E. PLINIAN- (or Vesuvian) are a type of volcanic eruption, named for the historical eruption of Mount Vesuvius in 79 AD that buried the Roman towns of Pompeii and Herculaneum and, specifically, for its chronicler Pliny the Younger. The process powering Plinian eruptions starts in the magma chamber, where dissolved volatile gases are stored in the magma. The gases vesiculate and accumulate as they rise through the magma conduit. These bubbles agglutinate and once they reach a certain size (about 75% of the total volume of the magma conduit) they explode. The narrow confines of the conduit force the gases and associated magma up, forming an eruptive column. Eruption velocity is controlled by the gas contents of the column, and low-strength surface rocks commonly crack under the pressure of the eruption, forming a flared outgoing structure that pushes the gases even faster.

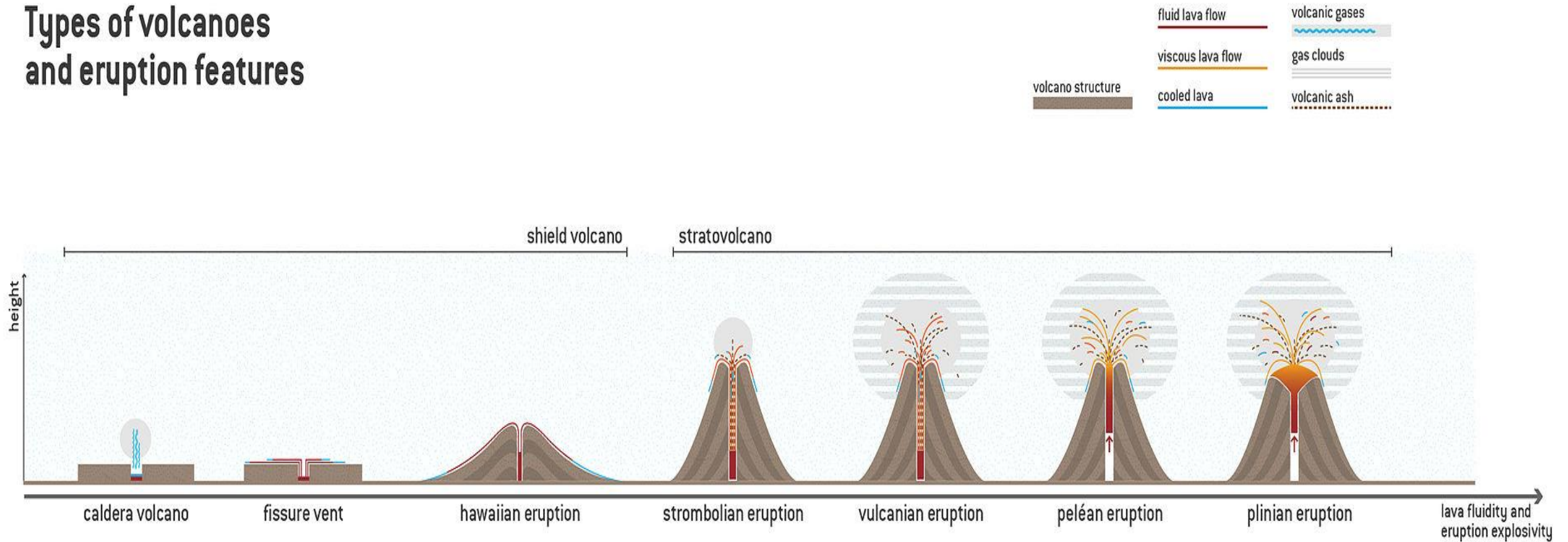
PLINIAN ERUPTION

Major Plinian eruptive events include:

- The AD 79 eruption of Mount Vesuvius buried the Roman towns of Pompeii and Herculaneum under a layer of ash and tephra. It is the model Plinian eruption. Mount Vesuvius has erupted several times since then. Its last eruption was in 1944 and caused problems for the allied armies as they advanced through Italy. It was the report by Pliny the Younger that lead scientists to refer to Vesuvian eruptions as "Plinian".
- The 1980 eruption of Mount St. Helens in Washington, which ripped apart the volcano's summit, was a Plinian eruption of Volcanic Explosivity Index (VEI) 5.
- The strongest types of eruptions, with a VEI of 8, are so-called "Ultra-Plinian" eruptions, such as the most recent one at Lake Toba 74 thousand years ago, which put out 2800 times the material erupted by Mount St. Helens in 1980.
- Hekla in Iceland, an example of basaltic Plinian volcanism being its 1947–48 eruption. The past 800 years have been a pattern of violent initial eruptions of pumice followed by prolonged extrusion of basaltic lava from the lower part of the volcano.
- Pinatubo in the Philippines on 15 June 1991, which produced 5 km³ (1 cu mi) of dacitic magma, a 40 km (25 mi) high eruption column, and released 17 megatons of sulfur dioxide.

PLINIAN ERUPTION

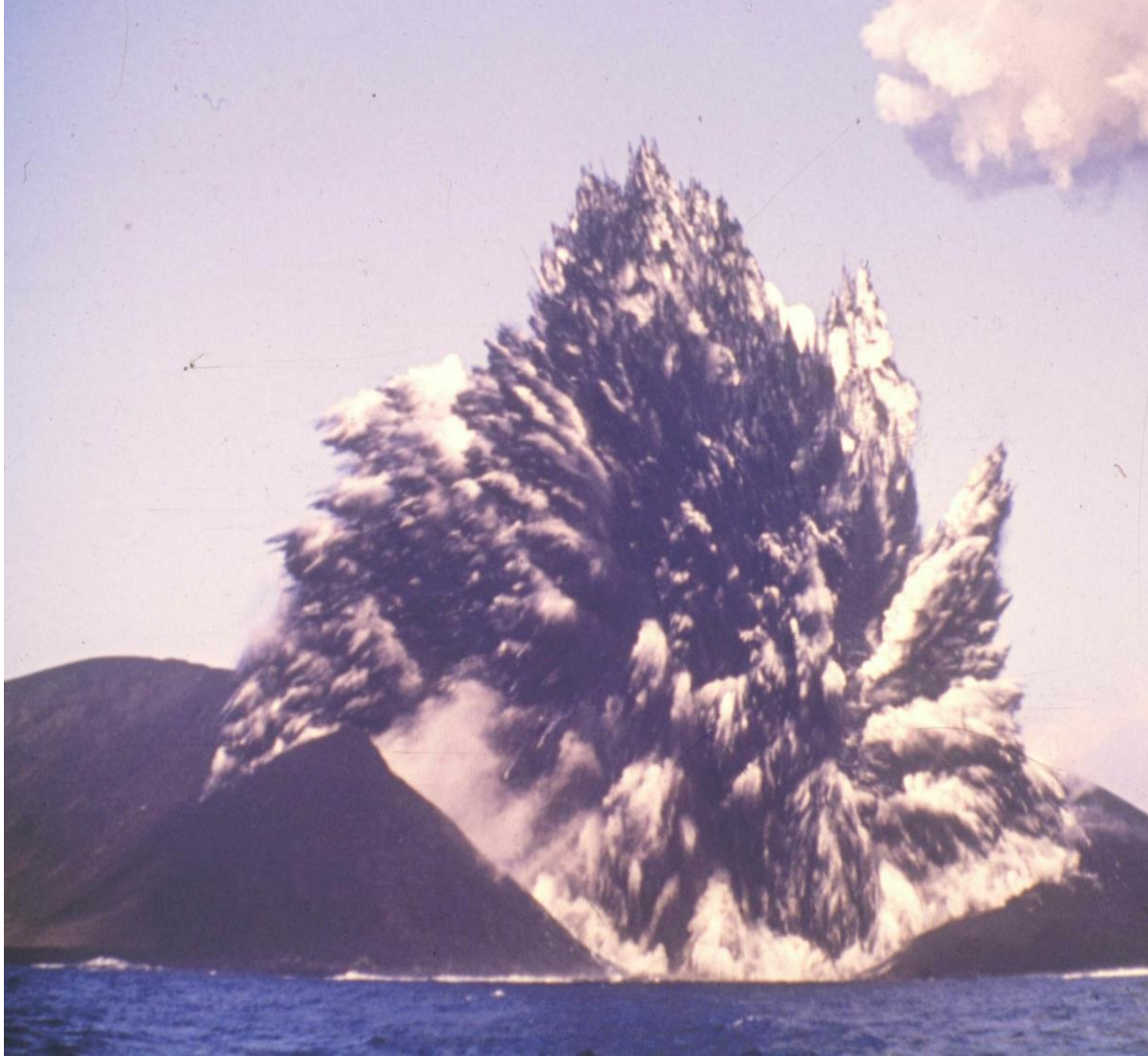
Types of volcanoes and eruption features



2. PHREATOMAGMATIC

SUBTYPES OF PHREATOMAGMATIC ERUPTION

A. SURTSEYAN- a type of volcanic eruption caused by shallow-water interactions between water and lava, named so after its most famous example, the eruption and formation of the island of Surtsey off the coast of Iceland in 1963. Surtseyan eruptions are the "wet" equivalent of ground-based Strombolian eruptions, but because of where they are taking place they are much more explosive. This is because as water is heated by lava, it flashes in steam and expands violently, fragmenting the magma it is in contact with into fine-grained ash. Surtseyan eruptions are the hallmark of shallow-water volcanic oceanic islands, however they are not specifically confined to them. Surtseyan eruptions can happen on land as well, and are caused by rising magma that comes into contact with an aquifer (water-bearing rock formation) at shallow levels under the volcano. The products of Surtseyan eruptions are generally oxidized palagonite basalts (though andesitic eruptions do occur, albeit rarely), and like Strombolian eruptions Surtseyan eruptions are generally continuous or otherwise rhythmic.



SURTSEYAN ERUPTION

Volcanoes known to have Surtseyan activity include:

- **Surtsey, Iceland**. The volcano built itself up from depth and emerged above the Atlantic Ocean off the coast of Iceland in 1963. Initial hydrovolcanic were highly explosive, but as the volcano grew out rising lava started to interact less with the water and more with the air, until finally Surtseyan activity waned and became more Strombolian in character.
- **Ukinrek Maars** in Alaska, 1977, and **Capelinhos** in the Azores, 1957, both examples of above-water Surtseyan activity.
- **Mount Tarawera** in New Zealand erupted along a rift zone in 1886, killing 150 people.
- **Ferdinanda**, a seamount in the Mediterranean Sea, breached sea level in July 1831 and was the source of a dispute over sovereignty between Italy, France, and Great Britain. The volcano did not build tuff cones strongly enough to withstand erosion, and disappeared back below the waves soon after it appeared.
- The underwater volcano **Hunga Tonga** in Tonga breached sea level in 2009. Both of its vents exhibited Surtseyan activity for much of the time. It was also the site of an earlier eruption in May 1988.



2. PHREATOMAGMATIC

SUBTYPES OF PHREATOMAGMATIC ERUPTION

b. SUBMARINE- type of volcanic eruption that occurs underwater. Submarine eruptions may produce seamounts which may break the surface to form volcanic islands and island chains. Submarine volcanism is driven by various processes. Volcanoes near plate boundaries and mid-ocean ridges are built by the decompression melting of mantle rock that rises on an upwelling portion of a convection cell to the crustal surface. Eruptions associated with subducting zones, meanwhile, are driven by subducting plates that add volatiles to the rising plate, lowering its melting point. Each process generates different rock; mid-ocean ridge volcanics are primarily basaltic, whereas subduction flows are mostly calc-alkaline, and more explosive and viscous

VOLCANIC HAZARDS

Pyroclastic Flow: A Deadly Force of Nature

Some of the most destructive forces of nature are Pyroclastic flows, which consist of hot gases infused with incandescent ash and larger lava fragments. Also referred to as *nuée ardentes* (glowing avalanches), these fiery flows are capable of racing down steep volcanic slopes at speeds that can exceed 100 kilometers (60 miles) per hour. A pyroclastic flow is composed of two parts—a low-density cloud of hot expanding gases containing fine ash particles and a ground-hugging portion that is often composed of pumice and other vesicular pyroclastic material.

Pyroclastic flows

Pyroclastic flows are composed of hot ash and pumice and/or blocky lava fragments that race down the slopes of volcanoes.



VOLCANIC HAZARDS

Lahars: Mudflows on Active and Inactive Cones

In addition to having violent eruptions, large composite cones may generate a type of fluid mudflow referred to by its Indonesian name, lahar. These destructive flows occur when volcanic debris becomes saturated with water and rapidly moves down steep volcanic slopes, generally following stream valleys. Some lahars are triggered when magma nears the surface of a glacially clad volcano, causing large volumes of ice and snow to melt. Others are generated when heavy rains saturate weathered volcanic deposits. Thus, lahars may occur even when a volcano is not erupting.

Lahars

A. This lahar raced down the Toutle River, which flows westward from Mount St. Helens, following the May 18, 1980, eruption.

B. Note the muddy consistency of lahar deposits.





Volcanic ash can destroy plant life and be drawn into the lungs of humans and other animals.



Ash and other pyroclastic materials can collapse roofs, or completely cover buildings.



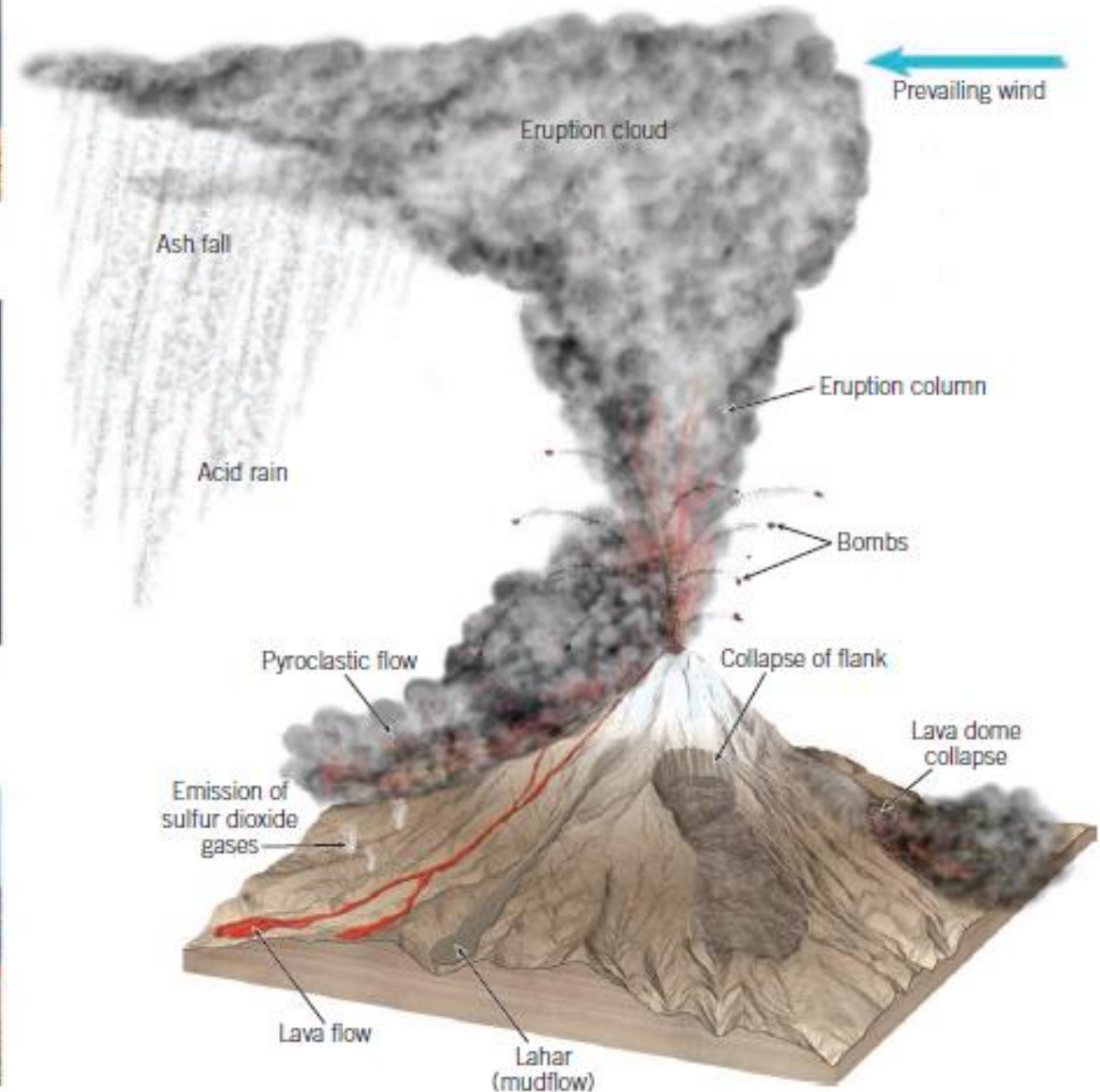
Lava flows can destroy homes, roads, and other structures in their paths.

Volcanic hazards

In addition to generating pyroclastic flows and lahars, volcanoes can be hazardous to human health and property, in many other ways.

FIGURE 5.17 Volcanic hazards

In addition to generating pyroclastic flows and lahars, volcanoes can be hazardous to human health and property, in many other ways.



- Reference:
Tarbuck, Lutgens and Tasa (2014), Earth: An Introduction to Physical Geology 11th ed., Pearson Education, Inc